

HONGYU HÈ

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(Version: September 2026 | 📄 indicates a hyperlink)

EDUCATION

Princeton University — <i>PhD in Electrical Engineering</i>	Aug. 2024 – present
▶ Committee: Maria Apostolaki [📄] (advisor), Sharad Malik [📄] , Aarti Gupta [📄]	United States
▶ GPA: 3.95/4.0 (8 graduate-level courses in EE and CS)	
ETH Zürich — <i>MSc in Computer Science</i>	Aug. 2021 – Jul. 2024
▶ Direct Doctoral Scholarship [📄] (5 out of 230 CS master's students in 2022)	Switzerland
▶ Thesis awarded highest grade: 6.0/6.0 (Advisor: Gustavo Alonso [📄])	
▶ Associated Researcher [📄] at ETH AI Center [📄]	
Vrije Universiteit Amsterdam — <i>BSc (Hons) in Computer Science</i>	Aug. 2018 – Jul. 2021
Universiteit van Amsterdam — <i>BSc Minor in Mathematics</i>	Sep. 2019 – Jan. 2021
▶ GPA rank: 1st of 180 [📄] ; national top 8 in CS [📄] (KHMW [📄] 2020)	The Netherlands
Tianjin University — <i>BA in Law</i> (unfinished)	Sep. 2016 – Aug. 2017
▶ Top 0.2% among 153,276 regional liberal arts students in National College Entrance Exam, “Gaokao” [📄]	China

INDUSTRY EXPERIENCE*

Microsoft[📄] — <i>Research Intern</i> Networking Research Group	Jun. 2026 – Sep. 2026
Curieo AI[📄] (sold) — <i>ML Research Fellow</i> remote, part-time	Jul. 2024 – Oct. 2024
IBM Research[📄] — <i>Engineering Intern</i> Geodata Team part-time	Apr. 2024 – Jun. 2024
Apple Inc.[📄] — <i>AI/ML Intern</i> Visual Intelligence Team (return offer)	May 2023 – Oct. 2023
Oracle Labs[📄] — <i>Graal Cloud Native Intern</i> Serverless Team part-time (return-intern offer)	Feb. 2023 – May 2023
Dexter Energy Services[📄] — <i>BSc Thesis Intern</i>	Jun. 2021 – Sep. 2021
Huawei Technologies[📄] (Amsterdam Research Center) — <i>R&D Intern</i> Search Team	Mar. 2021 – Jun. 2021
Picnic Technologies[📄] — <i>Software Developer</i> Store Team part-time	Sep. 2020 – Mar. 2021
— <i>Backend Engineer Intern</i> Mentor: Sander Mak [📄] (return offer)	Jun. 2020 – Sep. 2020

TEACHING

Princeton University 20 hrs/wk (head TA) Computer Networks [📄]	2025
ETH Zürich 14 hrs/wk Distributed Systems Lab, Big Data (my session recordings [📄])	2023, 2024
Vrije Universiteit Amsterdam 8–16 hrs/wk Graph Theory, Linear Algebra, C++, Calculus I	2019, 2020

TECHNICAL SKILLS

Programming & Scripting: C/C++, Python, (System)Verilog, Golang, Shell, Scala, Java, SQL, $\LaTeX$
Software & Tools: Kubernetes, Xilinx Vivado/Vitis suite, JAX, PyTorch, Linux, Git, Docker, Knative, Apache Spark, SLURM, STM32 (ARM Cortex-M3), MongoDB, PostgreSQL, Hadoop, HBase, Spring5, Qt5

*Industry recommendations: hhy.ee.princeton.edu/rec/[📄]

PUBLICATIONS

Preprint

- [14] H. Hè, M. Apostolaki. *Let AI Agents Translate Networks, Not Reason About Them*. [PDF[□]]

Peer-reviewed paper

- [13] H. Hè, M. Jin, M. Apostolaki. *Worst-Case Discovery and Runtime Protection for RL-Based Network Controllers*. USENIX NSDI 2027 (to appear) [PDF[□]]
- [12] H. Hè^{*}, A. Krentsel^{*}, S. Ratnasamy, M. Apostolaki. *Invariant Discovery for Networked Systems*. ACM HotNets 2026 (to appear) [PDF[□]]
- [11] C. Constantine, H. Hè, M. Apostolaki. *No Hyperscaler? No Problem. Your Students also Break (and Fix) Networks*. ACM HotNets 2026 (to appear)
- [10] M. Jin, H. Hè, M. Apostolaki. *Cross-Flow Correlations Survive Synthesis: Measuring Source-Level Privacy Leakage in Synthetic Network Traces*. NDSS 2027 (to appear) [PDF[□]]
- [9] H. Hè, M. Jin, M. Apostolaki. *Making Logic a First-Class Citizen in Generative ML for Networking*. USENIX NSDI 2026 [code[□]] [PDF[□]]
- [8] H. Hè, M. Apostolaki. *Just-in-Time Logic Enforcement: A new paradigm of combining statistical and symbolic reasoning for network management*. ACM HotNets 2025 [code[□]] [PDF[□]]
- [7] A. Zhou, C. Costic, H. Hè, A. Ghalayini, A. Kabbani, M. Apostolaki. *Mitigating Inter-datacenter Incast with a Proxy: The shortest path is not necessarily the fastest*. ACM HotNets 2025 [PDF[□]]
- [6] B. McKinzie, Z. Gan, J. Fauconnier, ..., H. Hè, ..., P. Gräsch, A. Toshev, Y. Yang. *MMI: Methods, Analysis & Insights from Multimodal LLM Pre-training*. ECCV 2024 **(first flagship multimodal LLM from 🍏)** [PDF[□]]
- [5] H. Hè, M. Friedman, T. Rekatsinas. *EnergAt: Fine-Grained Energy Attribution for Multi-Tenancy*. ACM HotCarbon 2023 **(adopted by Trycarbonara[□])** [media[□]] [code[□]] [PDF[□]]
- [4] B. Kruit, H. He, J. Urbani. *Tab2Know: Building a Knowledge Base from Tables in Scientific Papers*. ISWC 2020. [code[□]] [PDF[□]]

Patent

- [3] S. Kakarla, R. Beckett, H. Hè. *Autoformulation of Executable Cost-Saving Rules from Cloud Billing Policies with AI*. Under review.

Degree thesis

- [2] H. Hè. *FPGA-based SmartNIC for Distributed Machine Learning*. 2024 **(best grade: 6.0/6.0)** [MSc thesis[□]]
- [1] H. He. *How Can Datacenters Join the Smart Grid to Address the Climate Crisis? Using simulation to explore power and cost effects of direct participation in the energy market*. 2021 **(Amsterdam Thesis Award[□])** [BSc thesis[□]]

OTHER ACHIEVEMENTS

- ◆ Nominee for Princeton Teaching Award (3 nominees/department annually, see letters from my students here[□]) 2026
- ◆ Princeton Natural & Artificial Mind Fellowship in AI 2025
- ◆ Princeton University Administrative Fellowship in CS 2024
- ◆ Nominee for ETH Medal[□] (MSc) 2023
- ◆ Danish Government scholarship[□] [€ 28,260] 2021
- ◆ Leiden Excellence scholarship[□] [€ 15,000] 2021
- ◆ GRE 334 (verbal: 161, quant: 170, writing: 3) 2021
- ◆ GLOBE scholarship (ranked 1st out of 900+[□]) [€ 1,250] 2020
- ◆ ACM-ICPC: Amsterdam Algorithm Programming Preliminaries (AAPP) with my team, ranked 1st 2020
- ◆ ACM-ICPC: Benelux Algorithm Programming Contest (BAPC) with my team, ranked 9th 2019
- ◆ Young Talent Award[□] by *The Royal Holland Society of Sciences (KHMW)*[□] 2019
- ◆ Ranked 1st in Operating Systems (Cristiano Giuffrida[□]) and Networks (Andrew Tanenbaum[□]) courses 2019

LANGUAGES

English: TOEFL (speaking): 28, IELTS: 9 | **German:** B1 | **Dutch:** Basic | **Mandarin:** PSC[□]: Level 1-B, regional max.